

**Source: same
theorem for legs a b
and hypotenuse c**

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graph TD; A[Source: same theorem for legs a b and hypotenuse c] --> B[Assumption: nonnegative lengths]; B --> C[Inference: encode triangle constraint algebraically]; C --> D[Inference: expand and simplify]; D --> E[Inference: isolate c squared]; E --> F[Conclusion: a squared plus b squared equals c squared];
```

**Assumption:
nonnegative lengths**

**Inference: encode
triangle constraint
algebraically**

**Inference: expand
and simplify**

**Inference: isolate c
squared**

**Conclusion: a
squared plus b
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squared**